Ahmed Kheder

Cairo, Egypt ahmed.kheder2002@gmail.com +20 109 848 1767 ahmed-kheder1
AhmedKhadr1 AhmedKhadr

Profile

Computer and Systems Engineering graduate from Ain Shams University, ranked in the top 9% of the class. Focused on LLM agents and applied machine learning, working across the whole cycle: design, implementation, and measuring whether the result actually works. Comfortable moving between research ideas and clean, working code.

Education

Ain Shams University Cairo, Egypt
BSc in Computer and Systems Engineering *Sep 2021 – Jul 2026*

- **GPA:** 3.57/4.00 | 15th of 170 (Top 9%)
- **Key Modules:** Machine Learning, Deep Learning, Operating Systems, Database Systems, Data Structures & Algorithms, Advanced Programming, Artificial Intelligence, Fundamentals of Big-Data Analytics.

Experience

Nile University Giza, Egypt
Research Intern *Jul 2026 – Sep 2026*

- Researched short-horizon ECG forecasting, exploring reservoir computing, next-generation reservoir computing (NG-RC), Transformers, and time-series foundation models.

Graduation Project

AI-Driven Functional Verification for Digital Designs *In collaboration with Siemens*
Python, LangGraph, LangChain, MCP

- Built VAI, an autonomous LangGraph agent that tests chip designs, hitting 100% coverage on 4 of 5 open-source benchmarks and closing 84% of the testing gaps left by the traditional method
- Achieved full coverage in 7 minutes on average vs. 60-minute traditional runs, at ∼$0.30 LLM cost per design
- Designed the agent's core: a layered system prompt and an MCP-style tool layer with structured outputs
- Implemented tiered context management and a persistent cross-run memor

Projects

Real-Time ECG Waveform Forecasting | *Python, PyTorch, Chronos, Transformers, Fine-Tuning*

- Architected a streaming system that predicts the next second of ECG by splitting it into beat timing, type, shape, and a slow baseline, combining zero-shot Chronos, a beat-type classifier, and fine-tuned Chronos-Bolt models
- Recorded NMAE of 0.018 and MSE of 0.10 on unseen patients; outperformed the FADE paper (2025) on normal rhythm with 14% lower NMAE, 7% lower MAE, and 27% less shape distortion
- Devised a per-beat metric scoring timing, height, and shape, since standard metrics like MAE can reward flat, smeared forecasts; typical beats landed within 15 ms

Motor Imagery Classification | *Python, PyTorch, TensorFlow, MNE* *GitHub Repository*

- Automated Muse-S EEG preprocessing with MNE-Python: band-pass and notch filtering, ICA artifact removal, CAR, and z-score normalization
- Applied offline augmentation (Gaussian noise, temporal shifts, window jitter) and packaged data into efficient PyTorch Dataset/DataLoader pipelines
- Fine-tuned EEGNet-8-2 CNN and Bi-LSTM architectures in PyTorch/TensorFlow with reproducible training and checkpointing

San Francisco Crime Classification | *Python, Scikit-learn, LightGBM* *GitHub Repository*

- Engineered temporal and spatial features from 878K crime records, removed outliers and duplicates, and encoded/scaled inputs for 10-class classification
- Trained and tuned Random Forest, LightGBM, and Logistic Regression models; the best (Random Forest) scored 36% accuracy, 1.8× the 20% majority-class baseline, with log-loss 1.85 vs. 2.05 for Logistic Regression

Professional Development

Build AI Agents using MCP | *IBM, Coursera* | Apr 2026

Neural Networks and Deep Learning | *DeepLearning.AI, Coursera* | Sep 2025

Machine Learning Specialization (2 of 3 courses) | *DeepLearning.AI & Stanford Online* | Supervised Machine Learning; Unsupervised Learning, Recommenders, Reinforcement Learning

AI, Data Science & Machine Learning Diploma | *WE Telecom Egypt* | data cleaning, feature engineering, model tuning, deep learning with TensorFlow (CNNs, RNNs)

Python 3 Programming Specialization (4 of 5 courses) | *University of Michigan, Coursera*

Technical Skills

Languages: Python, C++, SQL

LLMs & Agents: LangGraph, LangChain, ReAct Agents, Prompt Engineering, Tool Calling / MCP, Context & Memory Management

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, MNE-Python

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Tools: Git/GitHub, MySQL, MongoDB